ALTEPLASE

High-alert drug

Strict eligibility checklist

CHEMICAL NAME Recombinant human tissue-type plasminogen activator (rt-PA), 527-amino-acid glycoprotein

CLASS Thrombolytic (fibrinolytic) agent

TRADE NAMES Actilyse · Activase · Cathflo (catheter clearance)

THERAPEUTIC USES / INDICATIONS

- Acute ischaemic stroke within 4.5 h of symptom onset (after CT excludes haemorrhage).
- Acute STEMI when primary PCI not available in time.
- Massive (high-risk) pulmonary embolism with haemodynamic instability.
- Occluded central venous catheters (low dose).

ADMINISTRATION

- **IV**: reconstitute with supplied water — swirl gently, do not shake.
- Bolus then infusion via pump; dedicated line.
- Avoid IM injections, arterial punctures, new central lines during and after.
- BP must be controlled before and during (stroke: <185/110 before, <180/105 for 24 h).

DOSAGES

- **Ischaemic stroke:** 0.9 mg/kg (max 90 mg) — 10% as IV bolus over 1 min, remaining 90% over 60 min.
- **STEMI (accelerated):** 15 mg bolus, 0.75 mg/kg over 30 min (max 50 mg), then 0.5 mg/kg over 60 min (max 35 mg).
- **Massive PE:** 10 mg bolus, then 90 mg over 2 h (or 0.6 mg/kg over 15 min in arrest).
- **Catheter clearance:** 2 mg instilled in lumen.

PHARMACOKINETICS

- **Onset:** immediate. Fibrin-specific.
- Cleared rapidly by liver.
- **t½:** 4–6 min (hence infusion).

PHARMACODYNAMICS

- Binds to **fibrin** in the clot and converts **plasminogen → plasmin** → lyses fibrin → dissolves thrombus.
- Relatively clot-specific, but systemic fibrinolysis still occurs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Minor bleeding at puncture sites, gums.
- Nausea, vomiting.
- Fever.

ADVERSE EFFECTS (SERIOUS)

- Intracranial haemorrhage (~6% in stroke).
- Major systemic bleeding.
- Orolingual angioedema (esp. with ACE inhibitors).
- Reperfusion arrhythmias; cholesterol embolism.

CONTRAINDICATIONS & CAUTIONS

- Any intracranial haemorrhage (current or past); stroke or head trauma within 3 months.
- Active internal bleeding; known bleeding diathesis; platelets <100,000; INR >1.7; DOAC within 48 h.
- Recent major surgery (<14 days); suspected aortic dissection.
- Uncontrolled BP >185/110 (stroke); intracranial neoplasm/AVM.

KEY INTERACTIONS

- **Anticoagulants, antiplatelets:** ↑ bleeding; avoid antithrombotics for 24 h after stroke thrombolysis (until repeat CT).
- **ACE inhibitors:** ↑ angioedema.

NURSING MANAGEMENT

- Complete eligibility checklist; confirm weight (estimate accurately) and time of onset.
- Neuro observations (NIHSS/GCS) and BP every 15 min for 2 h, every 30 min for 6 h, then hourly to 24 h (stroke).
- Stop infusion and call doctor for neuro deterioration, severe headache, vomiting, new bleeding.
- Watch tongue/lips for swelling.
- No NG tube, catheter or arterial lines for 24 h unless essential.

PATIENT EDUCATION

- Explain benefits and bleeding risk (family consent where required).
- Report headache, weakness, bleeding or swelling of tongue immediately.

KEY POINTS

- rt-PA; plasminogen to plasmin.
- ICH risk — strict checklist and BP control.
- Stroke: 0.9 mg/kg (max 90 mg) within 4.5 h, 10% bolus.
- Neuro obs every 15 min; angioedema with ACEi.